[Assignment] British Auction in RFQ System

1. Background
2. Problem Statement
3. Product Goals & Business Objectives
4. Success Metrics
5. Core Functional Requirements
6. British Auction Configuration Options
7. Validation Rules
8. Auction Listing & Details Pages
9. Deliverables

---

1. Background

What is an RFQ?

An **RFQ (Request for Quotation)** is a process where a buyer asks multiple suppliers to submit quotes for providing a service or product. Suppliers compete with each other by offering better prices and terms, and the buyer selects the most suitable quote.

What is a British Auction?

A **British Auction** (in this context) is a competitive bidding process where:

- Suppliers submit bids openly.
- Suppliers can continuously lower their prices to beat competitors.
- If bidding activity happens close to the auction end time, the auction is **automatically extended**.
- The auction has a **forced close time**, after which bidding must stop no matter what.

This approach prevents last-second bidding and encourages fair competition.

---

2. Problem Statement

Design and build a simplified RFQ system that supports **British Auction–style bidding**, including:

- Automatic bid-time extensions
- Forced close rules
- Configurable auction behavior
- Clear visibility of auction progress

---

3. Product Goals & Business Objectives

- Enable transparent and fair supplier competition
- Prevent last-minute bidding manipulation
- Encourage active participation from suppliers
- Help buyers achieve better final pricing

---

4. Success Metrics

- Increase in number of bids per RFQ
- Improved final prices compared to non-extended auctions

---

5. Core Functional Requirements

RFQ Creation

Users should be able to create an RFQ with the **British Auction** option enabled.

Key fields in the RFQ creation form:

- RFQ Name / Reference ID
- Bid Start Date & Time
- Bid Close Date & Time
- **Forced Bid Close Date & Time** (must be later than Bid Close Time)
- Pickup / Service Date

Example fields present in Quote Submission form for a RFQ:

- Carrier Name
- Freight charges , Origin charges , destination charges
- Transit Time
- Validity of Quote

> **Rule:** The auction must never extend beyond the Forced Bid Close Time.

---

6. British Auction Configuration Options

These configurations control **when** and **how** the auction time is extended.

6.1 Trigger Window (X Minutes)

Defines how close to the auction end the system should monitor bidding activity.

Example:

- Bid Close Time: 6:00 PM
- Trigger Window (X): 10 minutes
- The system monitors activity between **5:50 PM – 6:00 PM**

6.2 Extension Duration (Y Minutes)

Defines how much extra time should be added when a trigger condition occurs.

Example:

- Extension Duration (Y): 5 minutes
- If triggered, Bid Close Time extends from **6:00 PM → 6:05 PM**

6.3 Extension Triggers

Defines *what type of activity* should cause the auction to extend.

a) Bid Received in Last X Minutes

If **any supplier submits a new bid** during the trigger window, the auction extends.

Example:

- Trigger Window: 10 minutes
- A bid is placed at 5:55 PM
- Auction extends by Y minutes

---

b) Any Supplier Rank Change in Last X Minutes

If **any change in supplier ranking** occurs during the trigger window, the auction extends.

Example:

- Supplier B underbids Supplier A at 5:58 PM
- Rankings change
- Auction extends by Y minutes

---

c) Lowest Bidder (L1) Rank Change in Last X Minutes

The auction extends **only when the lowest-priced supplier changes**.

Example:

- Supplier C becomes the new lowest bidder at 5:59 PM
- Auction extends by Y minutes

---

7. Validation Rules

- Forced Bid Close Time must always be **greater than** Bid Close Time
- Auction extensions must **never exceed** the Forced Bid Close Time

---

8. Auction Listing & Details Pages

Auction Listing Page

Show a list of all British Auctions with:

- RFQ Name / ID
- Current Lowest Bid
- Current Bid Close Time
- Forced Close Time
- Auction Status (Active / Closed / Force Closed)

Auction Details Page

For a selected RFQ, display:

- All supplier bids (sorted by price)
- Supplier ranking (L1, L2, L3, ...)
- Details of the submitted quote like charges , quote validity ...
- Auction configuration (X and Y values)
- Activity log showing:
  - Bid submissions
  - Time extensions
  - Reason for each extension

---

9. Deliverables

You have to submit the following:

- Simple HLD with architecture diagram
- Schema Design for database tables
- Backend Code
- Frontend Code

Good luck and happy hacking!